Sagnik Bhattacharjee

Email | LinkedIn | GitHub

Education

Indian Institute of Engineering Science and Technology, Shibpur — Kolkata, India
BTech in Information Technology — *Expected July 2029*

Projects

amine | *Python, Flask, PyAutoGUI, SQLite* | GitHub | ProductHunt — August 2024 – October 2024

- Built distraction-blocking Pomodoro app with multi-threading for mouse boundary enforcement, keyboard blocking, and full-screen tracking, reaching #10 on ProductHunt with 450+ downloads
- Implemented system-level controls using PyAutoGUI and keyboard libraries to block Windows keys and manage application windows, preventing app switching during focus sessions unlike browser-based blockers
- Created analytics dashboard with Matplotlib, Seaborn, and Plotly for session data visualization and CSV export
- Developed session management system with SQLite database logging, URL validation, and desktop notifications, gaining 170+ GitHub stars

yars | *Python, requests, urllib3* | GitHub — October 2024

- Built Reddit scraping client with automatic retries, exponential backoff, proxy rotation, and randomized user agents to bypass rate limiting and anti-bot measures
- Implemented Reddit search functionality with global and subreddit-specific queries, handling pagination tokens and returning structured data including titles, URLs, and metadata
- Developed recursive comment extraction system parsing nested threads into hierarchical JSON structure
- Created user activity and subreddit feed scrapers supporting multiple sorting categories with request throttling

dunefetch | *Python, curses* | GitHub — August 2025

- Developed hybrid terminal application combining neofetch-style system information display with interactive falling sand physics simulation
- Engineered cellular automata-based particle system featuring 8 material types (sand, water, fire, oil, steam, plant, mud, wood) with realistic interaction physics
- Implemented terminal visualization using curses library with real-time particle updates, custom color schemes, and grid-based rendering
- Built modular CLI interface with keyboard controls for particle placement and system monitoring, using object-oriented design for extensible material physics

grudgepixel | *C++, Raylib, STL Containers* | GitHub — December 2021

- Built pixel art editor supporting 8×8, 16×16, 32×32 canvases with dynamic grid scaling and mouse-based tools
- Implemented frame management system using nested STL vectors for animation sequences with variable FPS playback
- Added symmetrical mirroring, adjustable brush sizes, three-palette color switching, grid overlays, and navigation features between frames
- Created drag-and-drop file handling with ImageResizeNN for image import and batch PNG export

physics engine | *C++, Raylib* | GitHub — May 2025

- Built 2D physics engine with rigid/static bodies, gravity, force application, and semi-implicit Euler integration
- Implemented collision detection for circle bodies with normal/penetration calculation, impulse-based response, and positional correction
- Created world manager handling physics updates, gravity application, pairwise collision resolution, and off-screen body removal
- Developed interactive demo with mouse-spawned bodies and collision visualization

Skills

Languages: Python, C/C++, SQL
Frameworks and Libraries: Flask, FlaskWebGUI, Raylib, curses, PyAutoGUI, pandas, NumPy, Matplotlib, Plotly, Seaborn
Developer Tools: Git, SQLite